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Hydraulic cylinder and system with pressure intensification

Abstract

A pressure cylinder includes a working cylinder and an intensification cylinder that is divided by a separator block. A working piston is arranged in the working cylinder and connected to a working rod that extends to an end portion. An intensification piston and an intensification rod are arranged in the intensification cylinder. A pump is configured to provide a pressurized hydraulic fluid to the pressure cylinder. A fluid reservoir is configured to supply a hydraulic fluid to the pump. A first valve is configured to selectively regulate hydraulic fluid flow between the pressure cylinder and the fluid reservoir and the pump. A second valve is configured to selectively regulate fluid flow between a first valve and the advance intensification chamber. A controller is in communication with the first valve and the second valve. The controller is configured to coordinate movement of the working piston and the intensification piston.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a hydraulic cylinder and a system employing pressure intensification.

BACKGROUND

(2) Pneumatic intensification cylinders have been used for resistance welding and metal working applications, such as bending, clinching, coining, forming, piercing, riveting, and stamping. Two examples of prevalent pneumatic intensification cylinders are described in U.S. Pat. Nos. 3,875,365 and 4,099,436, which are embodied in Applicant's OHMA® cylinders. One drawback of such cylinders is it is expensive to continuously generate and supply compressed air, and not all manufacturing locations may be able to supply air at the necessary pressure for operation.

(3) Servoelectric actuators are being increasingly applied in place of all forms of pneumatic cylinders for these process applications because control integration is claimed to make setup and maintenance of the servoelectric equipment easier, programmability leads to a reduction of inventory by eliminating the need to provide actuators with specific work and retract stroke combinations, and programmable operation improves functional capability (e.g., programmable stroke and coordinated motion).

(4) Servoelectric actuators with the capability to provide sufficient thrust, generally incorporate a ball or roller screw to convert rotary motion to linear motion. The force generation consistency and dynamic response of such actuators is directly related to the technology and manufacturing precision employed. Precision ball screws and planetary roller screws are expensive. Such actuators are comparatively large, subject to mechanical performance limitations, and increase mechanical complexity while decreasing operational reliability. Linear servomotors are available, but they do not deliver equivalent or sufficient values of force.

(5) Servoelectric control systems need to optimize torque and speed. Sizing a general application servoelectric motor and control to provide both high speed and high torque performance, results in a system larger than would be necessary for torque or speed control alone. Initial motion is usually based on velocity control to maximize the closing or advancing speed. When the tooling engages the workpiece, the control has to switch over to torque control to perform the principal work. Then the control has to revert back to velocity control to maximize retracting speed. The execution of these control methodology changes adds programming and tuning complexity and interruptions in the servo motion profile.

SUMMARY

(6) In one exemplary embodiment, a hydraulic pressure cylinder system includes a pressure cylinder that includes a working cylinder and an intensification cylinder that is divided by a separator block. A working piston is arranged in the working cylinder and connected to a working rod that extends to an end portion. The working piston separates the working cylinder into an advance working chamber and a retract working chamber. An intensification piston and an intensification rod are arranged in the intensification cylinder. The intensification piston separates

the intensification cylinder into an advance intensification chamber and a retract intensification chamber. A pump is configured to provide a pressurized hydraulic fluid to the pressure cylinder. A fluid reservoir is configured to supply a hydraulic fluid to the pump. A first valve is configured to selectively regulate hydraulic fluid flow between the pressure cylinder and the fluid reservoir and the pump. A second valve is configured to selectively regulate fluid flow between a first valve and the advance intensification chamber. A controller is in communication with the first valve and the second valve. The controller is configured to coordinate movement of the working piston and the intensification piston between a rest state, an advancing state, an advanced state, an intensified state, and a retracted state. The advanced state is configured to engage the end portion with a workpiece, and the intensified state is configured to perform an operation on the workpiece with the end portion.

(7) In a further embodiment of any of the above, the system includes a supply passage that fluidly connects the fluid reservoir to a supply side of the pump, a pressurized fluid passage that fluidly connects a pressure side of the pump to the first valve, a return passage that fluidly connects the first valve to a fluid reservoir, an advance passage that fluidly connects the first valve to the advance working chamber, and a retract passage that fluidly connects the first valve to the retract working chamber and the retract intensification chamber.

(8) In a further embodiment of any of the above, the first valve is a 4-way valve that includes a first position that is configured to fluidly connect the pressurized fluid passage to the return passage, a second position that is configured to fluidly connect the pressurized fluid passage to the advance passage, and to fluidly connect the retract passage to the return passage, and a third position that is configured to fluidly connect the advance passage to the return passage, and to fluidly connect the pressurized fluid passage to the retract passage.

(9) In a further embodiment of any of the above, the first valve is mounted to the fluid reservoir.

(10) In a further embodiment of any of the above, the second valve is a directional valve that includes a first position that fluidly blocks fluid flow from the advance passage to the advance intensification chamber, and a second position that fluidly connects the advance passage to the advance intensification chamber.

(11) In a further embodiment of any of the above, the second valve is mounted to the pressure cylinder.

(12) In a further embodiment of any of the above, the system includes an equalization circuit that fluidly connects the intensification passage to the retract passage. The equalization circuit includes a check valve that is configured to block fluid flow from the retract passage to the intensification passage but permit fluid flow from the intensification passage to the retract passage via a flow metering orifice.

(13) In a further embodiment of any of the above, the equalization circuit is mounted to the pressure cylinder.

(14) In a further embodiment of any of the above, the rest state includes the first valve in the first position, and the working piston and the intensification piston are both in retracted positions.

(15) In a further embodiment of any of the above, the advancing state includes the first valve in the second position and the second valve in the first position, with pressurized fluid that is configured to axially move the intensification rod toward the advance working chamber. The advanced state includes the first valve in the second position and the second valve in the first position, with the pressurized fluid that is configured to continue to axially move the intensification rod toward the advance working chamber until the flow of pressurized fluid to at least the advance working chamber is blocked by the intensification rod.

(16) In a further embodiment of any of the above, the intensified state includes the first valve in the second position and the second valve in the second position, wherein the intensification rod impinges into the advance working chamber.

(17) In a further embodiment of any of the above, the retracted state includes the first valve in the

third position and the second valve in the first position, with pressurized fluid that is configured to flow to the retract intensification chamber and the retract work chamber.

(18) In a further embodiment of any of the above, the system includes a pressure relief circuit that fluidly connects the pressurized fluid passage to the return passage. The pressure relief circuit is configured to normally block fluid flow from the pressurized fluid passage to the return passage but permit fluid flow from the pressurized fluid passage to the return passage above a predetermined pressure threshold.

(19) In a further embodiment of any of the above, the pressure relief circuit is mounted to the fluid reservoir.

(20) In a further embodiment of any of the above, the separator block includes a bore in which the intensification rod is at least partially arranged. A port is arranged on the separator block and fluidly connects the advance passage to the bore. The port is configured to be fluidly unblocked by the intensification rod in the rest state, the advancing state, the advanced state, and the retracted state. The port is configured to be fluidly blocked by the intensification rod in the intensified state.

(21) In a further embodiment of any of the above, the working piston and the working rod include a hole that is configured to receive the intensification rod in the intensified state.

(22) In a further embodiment of any of the above, the system includes a pressure sensor that is in communication with the controller and in fluid communication with the pressurized fluid passage. The controller is configured to monitor a pressure within the pressurized fluid passage that corresponds to an operational state of the system.

(23) In a further embodiment of any of the above, the pump is a positive displacement pump that is coupled to an electric motor.

(24) In a further embodiment of any of the above, the system includes a position sensor that is in communication with the controller. The controller is configured to monitor a longitudinal position of the end portion with the position sensor.

(25) In a further embodiment of any of the above, the system includes an air source that fluidly connects to the second valve by another valve. The other valve is configured to selectively provide pressurized air to the advance intensification chamber via the second valve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be further understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
- (2) FIG. 1 is a schematic view of an example robotic cell having a work station with a hydraulic intensification cylinder according to this disclosure.
- (3) FIG. 2 is a perspective view of an example hydraulic intensification cylinder and pressurized fluid supply assembly for the work station.
- (4) FIG. 3 is a detailed schematic of a hydraulic intensification cylinder system.
- (5) FIGS. 4A-4E are simplified fluid schematics of rest, advancing, advanced, intensified, and retracted states of the system shown in FIGS. 2 and 3.
- (6) FIG. 5A is a front view of the pressurized fluid supply assembly shown in FIG. 2.
- (7) FIGS. 5B and 5C are respectively cross-sectional views taken along lines 5B-5B and 50-5C in FIG. 5A.
- (8) FIGS. 6A-6F are cross-sectional views of the hydraulic intensification cylinder shown in FIG. 2 respectively illustrating retracted, advanced, intensified, high pressure stroke to complete work, return intensification piston to port open, and retracted positions, respectively.
- (9) FIG. 7 illustrates a fluid schematic in which a pneumatic intensification feature is employed.
- (10) The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the

following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible. Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(11) Referring to FIG. 1, a schematic of an example robotic cell **10** is illustrated. The cell **10** includes a first conveyor **12** that feeds a raw workpiece **14** into the cell **10** for one or more operations, such as bending, clinching, coining, forming, piercing, riveting, stamping, and/or resistance welding. A multi-axis robot **16** picks up the raw workpiece **14** and moves it into position at a first work station **18** for an operation. A first feeder **20** may supply a component to the work station, such as a rivet, fastener or other component that may be secured to the raw workpiece **14**. Once the operation is completed, the robot **16** may transfer the workpiece to another work station **22** that receives components from another feeder **24**. A fewer or greater number of work stations may be provided in the cell **10** than illustrated depending upon the desired operations. Once the operations have been completed on the workpiece, the robot **16** transfers the finished workpiece **28** to another conveyor **26** for processing outside of the cell **10** (e.g., to another manufacturing location or packaging of the finished part).

(12) In this disclosure, as shown in FIG. 2, one of the work stations **18** incorporates a pressure cylinder **30**, or hydraulic pressure cylinder, generally, that is in fluid communication with a pressurized hydraulic fluid supply assembly **32**, both of which may be mounted to a common frame **34** such as a pedestal. In a resistance welding application, the pressure cylinder **30** may instead be mounted to, or within, the resistance welding gun mounted to the common frame **34**.

(13) In one example, the pressure cylinder **30** includes a working cylinder **36** and an intensification cylinder **38** that are coaxially mounted with respect to one another by first and second end blocks **40**, **42** and a separator block **44** clamped together by fasteners **46**. It should be understood that the working cylinder **36** and intensification cylinder **38** may be decoupled from one another for packaging, if desired. As shown in FIGS. 4A-4E, the working cylinder **36** houses a working piston **92** from which a working rod **94** extends through the first end block **40** to an end portion **50**. The end portion **50** translates longitudinally during operation to cooperate with the workpiece. A tool, such as a spot or projection welding apparatus, a punch, or a die, is secured to the end portion **50** to engage and perform the operation on the workpiece.

(14) As shown in FIGS. 4A-4E and 6A-6F, the working piston **92** separates the working cylinder **36** into an advance working chamber **100** and a retract working chamber **102**. The intensification cylinder **38** houses an intensification piston **96** from which an intensification rod **98** extends at least partially into the separator block **44**. The intensification piston **96** separates the intensification cylinder **38** into an advance intensification chamber **104** and a retract intensification chamber **106**. It should be understood that the pressure cylinder **30** includes various seals, which are not shown, but known in the cylinder art.

(15) As shown in FIGS. 6A-6F, the intensification rod **98** extends through seal **118** and into the separator block **44**. When intensification, or amplification of the pressure in the advance working chamber **100** is desired, the intensification rod **98** is advanced through seal **120** to prevent fluid from escaping the advance working chamber **100**. At that point, pressure applied to the intensification piston **96** acts upon the smaller diameter of the intensification rod **98** causing it to raise the pressure in the advance working chamber **100**, with the pressure increase proportional to the ratio of the areas of the intensification piston **96** and intensification rod **98**. Because the volume of fluid displaced by the intensification rod **98** is comparatively small when compared to the volume of the advance working chamber **100**, the intensification piston **96** has to stroke a proportionally long distance to cause displacement of the working rod **94**.

(16) In situations where the advance stroke of the working rod **94** suddenly becomes unconstrained, such as when the tooling attached to the end portion **50** has completed its function (e.g., piercing or

shearing application where the material suddenly yields to the tool), the pressure in advance working chamber **100** will suddenly drop. This eliminates the back pressure acting against the intensification rod **98** and creates the possibility that the intensification piston **96** might lunge forward to the limit of its stroke, with the intensification piston **96** impacting the separator block **44**. Enhanced hydraulic positioning control of the intensification piston **96** provides the ability to prevent such impact and also to proceed through the operating sequence as soon as the completion of the process has been detected.

(17) Returning to FIGS. **2** and **3**, hydraulic fluid is communicated to the pressure cylinder **30** via ports **48a-48f** (illustrated with hydraulic fittings in place), which are fluidly connected to the pressurized fluid supply assembly **32** as shown in the schematics provided in FIGS. **3-4E**. The pressurized fluid supply assembly **32** includes a motor **54** (e.g., servomotor) rotationally driving a pump **56**, such as a positive displacement pump. Pump pressure sensing (e.g., pressure sensor **128** in FIG. **3**) could be added to minimize the pump run time and hydraulic fluid heating. The pump **56** driven by a servomotor provides more precise control of speed and position. Sensing could be provided to determine cylinder working rod extension or the position of the connected tool (e.g., linear sensor **126** in FIG. **3**). The pump pressure can have a wide range of 10 PSI to 300 PSI (or higher depending on the intensification ratio and seal pressure rating) as compared to a system operating from a typical shop air supply, enabling a reduced number of smaller cylinder bore sizes. An electric motor driven pump facilitates the control of position, speed and force. The system, with incompressible fluid, quickly and effectively blocks motion when power is removed, unlike a pneumatic cylinder which may be pulled from the returned position by gravity or move suddenly when compressed air is exhausted. The fail-safe control system with lockout capability is much less complicated to implement than one powered by compressed air since there are fewer components and potential failure modes. It is only necessary to remove electrical power to the motor **54**.

(18) The motor **54** and the pump **56** are mounted on a common bracket **58** that is secured to a first plate **60**, shown in FIG. **2**. A hydraulic fluid reservoir **64** is arranged between the first plate **60** and a second plate **62** secured to one another by fasteners **65**. A relief valve **66** and a first valve (e.g., 4-way 3-position valve) **68** are supported on the second plate **62**, which may include one or more internal passageways in the plate, as shown in FIGS. **5A-5C**. The first valve **68** is configured to selectively regulate hydraulic fluid flow between the working cylinder **36** or the intensification cylinder **38** and the fluid reservoir **32** and the pump **56**.

(19) Within the hydraulic fluid reservoir **64**, a baffle **114** is attached to the first plate **60** to cause fluid returning from the pump **56** to be redirected and diffused into the fluid within the reservoir. Similarly, a return extension **124** is connected to the second plate **62** to direct fluid returning through the first valve **68** so it will discharge below the surface of the hydraulic fluid in the reservoir. Baffle **114** and return extension **124** prevent the aeration of the hydraulic fluid, and the performance degradation that would result from entraining air in the otherwise incompressible hydraulic fluid. Examples are cavitation within the pump **56** and a reduction of the intensification efficiency in the pressure cylinder **30**.

(20) The disclosed system is filled with hydraulic fluid (i.e., oil), which enables the reservoir **64** to be smaller than a conventional intensification cylinder reservoir where sufficient additional hydraulic fluid must be provided as air is pushed out of the cylinder as the working piston advances. The reservoir **64** provides some extra make-up fluid, venting of entrained air released from the hydraulic fluid or system (e.g., via a breather **112**), and may provide visibility of the fluid level and condition (e.g., clarity and color). These features improve the operational efficiency when entrained air and contamination are introduced to the system through poor maintenance procedures. The reservoir **64** is designed to passively cool the fluid, which is heated during compression. The reservoir **64**, for example, may be constructed of aluminum with fins to increase the area for dissipating heat. The reservoir **64** also accommodates the transfer of hydraulic fluid from one side of a cylinder piston to the other, without requiring any supplemental methods (e.g., bladders,

spring-loaded follower) to account for the volume differences that arise from the fact the pressurized area is different on each side of the piston.

(21) With continuing reference to FIGS. 2 and 4A-4E, a supply passage **70** fluidly connects the fluid reservoir **64** to the pump **56** on the supply side of the pump **56**. A pressurized fluid passage **72** fluidly connects a pressure side of the pump **56** to the first valve **68**. A return passage **74** fluidly connects the first valve **68** to the fluid reservoir **64**. In one example, a pressure relief circuit **76**, which may be provide in the second plate **62** of the fluid reservoir **64**, fluidly interconnects the pressurized fluid passage to the return passage **74** that fluidly interconnects the first valve **68** to the fluid reservoir **64**. The pressure relief circuit **76** is configured to normally block fluid flow with a pressure relief valve **66** from the pressurized fluid passage **72** to the return passage **74** but permit fluid flow from the pressurized fluid passage **72** to the return passage **74** above a predetermined pressure threshold. The pressure relief valve **66** is provided as a safety device so the system cannot be over-pressurized to a value that will exceed the operating limit rating of the cylinder, the machine or gun frame, tooling, or the workpiece. In a less sophisticated control regime, this pressure relief valve could serve as an active control element, and used to determine the required working pressure.

(22) An advance passage **78** fluidly connects an advance port **108** (FIGS. 4A-4E and 5C) in communication with the first valve **68** to the advance working chamber **100** via port **48f**. Ports **48a**, **48b** are in fluid communication respectively with the retract working chamber **102**, and the retract intensification chamber **106**.

(23) The first valve **68** includes a first position **68a** configured to fluidly connect the pressurized fluid passage **72** to the return passage **74**. A second position **68b** is configured to fluidly connect the pressured fluid passage **72** to the advance passage **78**, and to fluidly connect the retract passage **80** to the return passage **74**. A third position **68c** is configured to fluidly connect the advance passage **78** to the relief passage, and to fluidly connect the pressurized fluid passage to the retract passage.

(24) In the example, a directional control valve (i.e., second valve) **52**, is mounted to the second end block **42** of the pressure cylinder **30**. The second valve/directional control valve **52** is configured to selectively regulate fluid flow between first valve **68** and the advance intensification chamber **104**. An intensification passage **84** fluidly connects the directional valve **52** to the advance intensification chamber **104**. The directional valve **52** includes a first position **52a** fluidly blocking fluid flow from the advance passage **78** to the advance intensification chamber **104**, and a second position **52b** fluidly connecting the advance passage **78** to the advance intensification chamber **104**. When the intensification rod **98** is retracted, the working rod **94** of pressure cylinder **30** may be stopped in any position by coordinating the first valve **68** and the pump **56**. When the desired working rod **94** position has been achieved, the first valve **68** can be returned to its first position **68a**, which would stop hydraulic fluid returning to the fluid reservoir **64**. With the hydraulic fluid blocked by the first valve **68**, the motor **54** and pump **56** would not need to be operated to maintain the working rod **94** position. This greatly reduces the proliferation of cylinder variations in a manufacturing facility because conventional air cylinders are mechanically configured to provide a specific combination of working stroke and retract stroke.

(25) A retract passage **80** fluidly connects a retract port **110** from the first valve **68** to the retract working chamber **102** and retract intensification chamber **106** respectively via ports **48a**, **48b**. An equalization circuit **86**, mounted to the pressure cylinder **30**, is fluidly connected between the intensification passage **84** and the retract passage **80**. In the example, port **48c** is provided at the junction between the equalization circuit **86** and the retract passage **80**, and port **48d** is provided in the advanced passage upstream from the directional valve **52**. The equalization circuit **86** includes a check valve **88** configured to block fluid flow from the retract passage **80** to the intensification passage **84** but permit fluid flow from the intensification passage **84** to the retract passage via a flow metering orifice **90**. The orifice **90** is active when the second valve **52** is open, to provide some return pressure against both working and intensifying pistons **92**, **96** to prevent piston drift.

The orifice **90** also provides a bypass for the fluid applied to the intensification piston **92** to establish a relationship between pump speed and intensification pressure.

(26) A controller **130** (FIG. **3**) is in communication with the first valve **68** and the second valve **52**. The controller **130** is configured to coordinate movement of the working piston **92** and the intensification piston **96** between a rest state (FIG. **4A**), an advancing state (FIG. **4B**), an advanced state (FIG. **4C**), an intensified state (FIG. **4D**), and a retracted state (FIG. **4E**). The advanced state is configured to engage the end portion **50**, with its tool (not shown), with the workpiece. The intensified state is configured to perform an operation on the workpiece with the end portion **50** and tool.

(27) With reference to FIG. **3**, the controller **130** may communicate with a position sensor **126** (e.g., LVDT) to monitor a longitudinal position of the end portion **50** to provide capability for analysis and verification of a process signature, such as a force-displacement or velocity-displacement profile. The controller **130** can selectively use a variety of input signals to control operation. For example, the position sensor **126** can provide position feedback during the advance stroke. Some examples, include providing precise control from the signal generated during the collapse of a resistance weld projection (e.g., U.S. Pat. No. 7,564,005 entitled “RESISTANCE WELDING FASTENER AND MONITOR AND METHOD OF USING SAME”, issued Jul. 21, 2009, and incorporated herein by reference in its entirety), piercing of a hole, or completion of a clinch (e.g., U.S. Provisional Patent Application No. 63/425,447 entitled “SELF-PIERCING CLINCH FASTENER INSTALLATION PRESS”, filed Nov. 15, 2022, and incorporated herein by reference in its entirety).

(28) The controller **130** may also communicate with a pressure sensor **128** that is in fluid communication with the pressurized fluid passage **72**. The controller **130** is configured to monitor a pressure within the pressurized fluid passage corresponding to an operational state of the system, for example, by sensing contact and verifying the operating pressure, and to associate pressure values with the position feedback. The fluid pressure sensor **128** can be used to identify unusual pressure decay during the rest state indicative of system leakage, including early detection of cylinder seal deterioration; if the pump **56** should be cycled periodically (a maintenance cycle) to remove excessive entrained air from the hydraulic fluid; if the fluid reservoir **64** has insufficient hydraulic fluid; unexpected pressure during the advance stroke indicating an obstacle has been encountered; the moment the tool has engaged the workpiece and the quality of contact or workpiece stability; the moment the tool has completed its task, since the pressure may build to indicate motion has ceased, or dropped to indicate the task (e.g., piercing a hole) has been completed; the moment an event has occurred, so cycle time may be optimized by immediately advancing to the next phase of the operating sequence; a detailed and sensitive process signature considering that the ratio of the intensification rod **98** and working piston **92** will proportionally amplify pressure changes at the working rod **94** for the pressure sensor, which is very useful for processes where pressure is a useful indicator of process quality such as metal forming, press-fit assembly, and broaching; excessive return force indicating the tooling has not disengaged from the workpiece; and/or unexpected pressure during the return stroke indicating an obstacle has been encountered.

(29) Other aspects of operation of the cell **10** may be coordinated by the controller **130**, such as movement of the robot **16**, various aspects of the first and second work stations **18**, **22**, and operation of the first and second feeders **20**, **24** and of the first and second conveyors **12**, **26**.

(30) In terms of hardware architecture, such a computing device can include a processor, memory, and one or more input and/or output (I/O) device interface(s) that are communicatively coupled via a local interface. The local interface can include, for example but not limited to, one or more buses and/or other wired or wireless connections. The local interface may have additional elements, which are omitted for simplicity, such as controllers, buffers (caches), drivers, repeaters, and receivers to enable communications. Further, the local interface may include address, control,

and/or data connections to enable appropriate communications among the aforementioned components.

(31) The controller **130** may be a hardware device for executing software, particularly software stored in memory. The controller **130** can be a custom made or commercially available processor, a central processing unit (CPU), an auxiliary processor among several processors associated with the controller, a semiconductor-based microprocessor (in the form of a microchip or chip set) or generally any device for executing software instructions.

(32) The memory can include any one or combination of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, VRAM, etc.)) and/or nonvolatile memory elements (e.g., ROM, hard drive, tape, CD-ROM, etc.). Moreover, the memory may incorporate electronic, magnetic, optical, and/or other types of storage media. The memory can also have a distributed architecture, where various components are situated remotely from one another, but can be accessed by the processor.

(33) The software in the memory may include one or more separate programs, each of which includes an ordered listing of executable instructions for implementing logical functions. A system component embodied as software may also be construed as a source program, executable program (object code), script, or any other entity comprising a set of instructions to be performed. When constructed as a source program, the program is translated via a compiler, assembler, interpreter, or the like, which may or may not be included within the memory.

(34) The disclosed input and output devices that may be coupled to system I/O interface(s) may include input devices, for example but not limited to, a keyboard, mouse, scanner, microphone, camera, mobile device, proximity device, etc. Further, the output devices, for example but not limited to, a printer, display, etc. Finally, the input and output devices may further include devices that communicate both as inputs and outputs, for instance but not limited to, a modulator/demodulator (modem; for accessing another device, system, or network), a radio frequency (RF) or other transceiver, a telephonic interface, a bridge, a router, etc.

(35) When the controller **130** is in operation, the processor can be configured to execute software stored within the memory, to communicate data to and from the memory, and to generally control operations of the computing device pursuant to the software. Software in memory, in whole or in part, is read by the processor, perhaps buffered within the processor, and then executed.

(36) Referring to FIGS. **4A**, **6A** and **6F**, the system is illustrated in a rest state, in which the working piston **92** and an intensification piston **96** are in the retracted positions. In this state, the first valve **68** is in a first position **68a**, which fluidly connects the pressurized fluid passage **72** to the return passage **74**. The directional valve **52** is in a first position **52a**.

(37) To actuate the pressure cylinder **30** to an advancing state (FIGS. **4B** and **6B**), the first valve **68** is moved to a second position **68b** in which the pressurized fluid passage **72** is fluidly connected to the advance passage **78**, and the retract passage **80** is fluidly connected to the return passage **74**. Pressurized fluid flows from the pump **56** and pressurized fluid passage **72** to advance passage **78** via the first valve **68** in its second position **68b** to provide pressurized fluid to the advance working chamber **100**, which advances the working piston **92** and its working rod **94** toward the workpiece **14** until the end portion **50** and its supported tool engages the workpiece **14**, as shown in FIG. **4C**.

(38) At this point, the directional valve **52** is actuated to a second position **52b**, which enables pressurized fluid flow from the advanced passage **78** to the advance intensification chamber **104** via the intensification passage **84**. The separator block **44** includes a bore **82** in which the intensification rod **98** is supported by and at least partially arranged. The port **48f** fluidly connects the advance passage **78** to the bore **82**.

(39) As the intensification rod **98** advances through the bore **82** in the separator block **44** and passes through seal **120** (FIG. **4C**), flow from the advanced passage **78** is prevented from further flow into the advance working chamber **100**. As the intensification rod **98** continues to advance into the advance working chamber **100** to the intensified state (FIGS. **4D** and **6C**), the force across the

working piston **92** and thus the working rod **94** significantly increases on the workpiece **14**. If desired, the working piston **92** (and, optionally, working rod **94**) include a hole **116** configured to receive the intensification rod **98** in the intensified state (FIG. **6D**).

(40) Once the operation is completed, the intensification piston **96** and its respective intensification rod **98** are moved to a position where the intensification rod **98** no longer penetrates seal **120**. At that point, advance working chamber **100** is once again connected to advanced passage **78**, permitting the fluid in retract working chamber **102** to push against working piston **92** to force the fluid from the advance working chamber **100**, back through the advance passage **78** and return passage **74**. As shown in FIG. **4E**, the first valve **68** is moved to a third position **68c** in which the advance working chamber **100** and advance intensification chamber **104** are exhausted through the first position **52a** of the directional valve **52**, through the advanced passage **78**, through the third position **68c** of the first valve **68**, into the return passage **74** and back to the fluid reservoir **64**. At the same time, a pressurized fluid from the pressurized fluid passage **72** is supplied through the third position **68c** of the first valve **68** to the retract passage **80** and into the retract working chamber **102** and retract intensification chamber **106**, which are at higher relative pressures than the advance working chamber **100** and the advance intensification chamber **104**.

(41) Referring to FIG. **7**, the intensification function can be operated using compressed air instead of hydraulic fluid for applications where the compressibility and viscosity of air provide desirable performance characteristics. In one example, a regulated air source **150** is fluidly connected to a 4-way 2-position second valve **152** by an isolation valve **158** via a supply passage **154** and an advance passage **178**. The at-rest second valve **152** is configured to provide air to **48b**, retaining the intensification piston **98** in the retracted position. After the working rod **94** has advanced to the work position, the second valve **152** is operated, to provide pressurized air to the advance intensification chamber **104** via an intensification passage **184**. In this method of operation, the low viscosity and compressibility of air may provide useful dynamic response for applications where it is beneficial to respond to variations in the load force (e.g., in welding applications where thermal expansion and workpiece upset could cause sudden changes in pressure). In this case, operation of the intensification cylinder **36** is similar to the prior art so it is easy to explain and execute.

Operation of the intensification cylinder **38** with air does not diminish the advantage of operating the pressure cylinder **30** using the pressurized hydraulic fluid supply assembly **32**. Hydraulic operation of the pressure cylinder **30** facilitates positive and accurate control over the working rod **94** position, employs hydraulic fluid on both sides of the working piston **92** seals to minimize the effects of any seal leakage, and retains the capability to achieve higher values of retract force.

(42) The intensification (amplification) ratio provides the capability to simply produce a range of output forces with a limited range of input pressure. In resistance welding, this is useful to perform tip dressing at a low force, welding at a high force, and forging at a higher force. While the pressure range is limited for an air system using shop supplied compressed air at 20 psi to 65 psi, the 10 psi to 300 psi range of hydraulic pressure supplied by pump **56** enables a broader output force range having high precision.

(43) The hydraulic amplification of the invention is highly efficient in transferring energy from the motor to the cylinder rod, and vice-versa. The disclosed arrangement employs a single hydraulic fluid pump and an elegant (simple and well structured) arrangement of hydraulic control elements to operate both the working and intensification functions. In addition to eliminating the requirement for compressed air, this high-thrust actuator system delivers servo-like control of working rod speed, position, and thrust.

(44) Configuration and tuning of the system for position, pressure, and torque control is easily accomplished because the load is a fairly simple pump that is loaded in principally one direction. Configuration and tuning scales the control parameters to provide the optimum cylinder response based on factors such as the cylinder bore size and the pump size. The result is a system that is similar to a transmission in its ability to efficiently deliver both high-speed low-force and low-

speed high-force output.

(45) It should also be understood that although a particular component arrangement is disclosed in the illustrated embodiment, other arrangements will benefit herefrom. Although particular step sequences are shown, described, and claimed, it should be understood that steps may be performed in any order, separated or combined unless otherwise indicated and will still benefit from the present invention.

(46) Although the different examples have specific components shown in the illustrations, embodiments of this invention are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples.

(47) Although an example embodiment has been disclosed, a worker of ordinary skill in this art would recognize that certain modifications would come within the scope of the claims. For that reason, the following claims should be studied to determine their true scope and content.

Claims

1. A hydraulic pressure cylinder system, comprising: a pressure cylinder including a working cylinder and an intensification cylinder divided by a separator block, a working piston arranged in the working cylinder and connected to a working rod extending to an end portion, the working piston separating the working cylinder into an advance working chamber and a retract working chamber, an intensification piston and an intensification rod arranged in the intensification cylinder, the intensification piston separating the intensification cylinder into an advance intensification chamber and a retract intensification chamber; a pump configured to provide a pressurized hydraulic fluid to the pressure cylinder; a fluid reservoir configured to supply a hydraulic fluid to the pump; a first valve configured to selectively regulate hydraulic fluid flow between the pump and at least the advance working chamber, the retract working chamber, and the retract intensification chamber; a second valve configured to selectively regulate fluid flow between the first valve and the advance intensification chamber; a controller in communication with the first valve and the second valve, the controller configured to coordinate movement of the working piston and the intensification piston between a rest state, an advancing state, an advanced state, an intensified state, and a retracted state, wherein the advanced state is configured to engage the end portion with a workpiece, and the intensified state is configured to perform an operation on the workpiece with the end portion; a supply passage fluidly connecting the fluid reservoir to a supply side of the pump; a pressurized fluid passage fluidly connecting a pressure side of the pump to the first valve; a return passage fluidly connecting the first valve to a fluid reservoir; an advance passage fluidly connecting the first valve to the advance working chamber; and a retract passage fluidly connecting the first valve to the retract working chamber and the retract intensification chamber; and wherein the separator block includes a bore in which the intensification rod is at least partially arranged, a port is arranged on the separator block and fluidly connects the advance passage to the bore, wherein the port is configured to be fluidly unblocked by the intensification rod in the rest state, the advancing state, the advanced state, and the retracted state, and wherein the port is configured to be fluidly blocked by the intensification rod in the intensified state.

2. The system of claim 1, wherein the first valve is a 4-way valve including: a first position is configured to fluidly connect the pressurized fluid passage to the return passage; a second position is configured to fluidly connect the pressurized fluid passage to the advance passage, and to fluidly connect the retract passage to the return passage; and a third position is configured to fluidly connect the advance passage to the return passage, and to fluidly connect the pressurized fluid passage to the retract passage.

3. The system of claim 2, wherein the first valve is mounted to the fluid reservoir.

4. The system of claim 2, wherein the second valve is a directional valve including: a first position

fluidly blocking fluid flow from the advance passage to the advance intensification chamber; and a second position fluidly connecting the advance passage to the advance intensification chamber.

5. The system of claim 4, wherein the second valve is mounted to the pressure cylinder.

6. The system of claim 4, wherein the rest state comprises the first valve in the first position, and the working piston and the intensification piston are both in retracted positions.

7. The system of claim 4, wherein the advancing state comprises the first valve in the second position and the second valve in the first position, with pressurized fluid configured to axially move the intensification rod toward the advance working chamber; and wherein the advanced state comprises the first valve in the second position and the second valve in the first position, with the pressurized fluid configured to continue to axially move the intensification rod toward the advance working chamber until the flow of pressurized fluid to at least the advance working chamber is blocked by the intensification rod.

8. The system of claim 4, wherein the intensified state comprises the first valve in the second position and the second valve in the second position, wherein the intensification rod impinges into the advance working chamber.

9. The system of claim 4, wherein the retracted state comprises the first valve in the third position and the second valve in the first position, with pressurized fluid configured to flow to the retract intensification chamber and the retract work chamber.

10. The system of claim 1, comprising a pressure relief circuit fluidly connecting the pressurized fluid passage to the return passage, the pressure relief circuit configured to normally block fluid flow from the pressurized fluid passage to the return passage but permit fluid flow from the pressurized fluid passage to the return passage above a predetermined pressure threshold.

11. The system of claim 10, wherein the pressure relief circuit is mounted to the fluid reservoir.

12. The system of claim 1, wherein the working piston and the working rod include a hole configured to receive the intensification rod in the intensified state.

13. The system of claim 1, comprising a pressure sensor in communication with the controller and in fluid communication with the pressurized fluid passage, the controller configured to monitor a pressure within the pressurized fluid passage corresponding to an operational state of the system.

14. The system of claim 1, wherein the pump is a positive displacement pump coupled to an electric motor.

15. The system of claim 1, comprising a position sensor in communication with the controller, the controller configured to monitor a longitudinal position of the end portion with the position sensor.

16. A hydraulic pressure cylinder system, comprising: a pressure cylinder including a working cylinder and an intensification cylinder divided by a separator block, a working piston arranged in the working cylinder and connected to a working rod extending to an end portion, the working piston separating the working cylinder into an advance working chamber and a retract working chamber, an intensification piston and an intensification rod arranged in the intensification cylinder, the intensification piston separating the intensification cylinder into an advance intensification chamber and a retract intensification chamber; a pump configured to provide a pressurized hydraulic fluid to the pressure cylinder; a fluid reservoir configured to supply a hydraulic fluid to the pump; a first valve configured to selectively regulate hydraulic fluid flow between the pressure cylinder and the fluid reservoir and the pump; a second valve configured to selectively regulate fluid flow between the first valve and the advance intensification chamber; and a controller in communication with the first valve and the second valve, the controller configured to coordinate movement of the working piston and the intensification piston between a rest state, an advancing state, an advanced state, an intensified state, and a retracted state, wherein the advanced state is configured to engage the end portion with a workpiece, and the intensified state is configured to perform an operation on the workpiece with the end portion; and an equalization circuit fluidly connecting an intensification passage to the retract passage, the equalization circuit including a check valve configured to block fluid flow from the retract passage to the intensification passage

but permit fluid flow from the intensification passage to the retract passage via a flow metering orifice.

17. The system of claim 16, wherein the equalization circuit is mounted to the pressure cylinder.

18. The system of claim 16, comprising: a supply passage fluidly connecting the fluid reservoir to a supply side of the pump; a pressurized fluid passage fluidly connecting a pressure side of the pump to the first valve; a return passage fluidly connecting the first valve to a fluid reservoir; an advance passage fluidly connecting the first valve to the advance working chamber; a retract passage fluidly connecting the first valve to the retract working chamber and the retract intensification chamber; wherein the first valve is a 4-way valve including: a first position is configured to fluidly connect the pressurized fluid passage to the return passage; a second position is configured to fluidly connect the pressurized fluid passage to the advance passage, and to fluidly connect the retract passage to the return passage; and a third position is configured to fluidly connect the advance passage to the return passage, and to fluidly connect the pressurized fluid passage to the retract passage; wherein the second valve is a directional valve including: a first position fluidly blocking fluid flow from the advance passage to the advance intensification chamber; and a second position fluidly connecting the advance passage to the advance intensification chamber.

19. A hydraulic pressure cylinder system, comprising: a pressure cylinder including a working cylinder and an intensification cylinder divided by a separator block, a working piston arranged in the working cylinder and connected to a working rod extending to an end portion, the working piston separating the working cylinder into an advance working chamber and a retract working chamber, an intensification piston and an intensification rod arranged in the intensification cylinder, the intensification piston separating the intensification cylinder into an advance intensification chamber and a retract intensification chamber; a pump configured to provide a pressurized hydraulic fluid to the pressure cylinder; a fluid reservoir configured to supply a hydraulic fluid to the pump; a first valve configured to selectively regulate hydraulic fluid flow the pump and the advance working chamber and the retract working chamber; an air source configured to provide pressurized air; a second valve fluidly connected to the second valve by another valve, the other valve configured to selectively provide the pressurized air to the second valve, wherein the second valve is configured to selectively regulate an airflow between the air source and the advance intensification chamber and the retract intensification chamber; and a controller in communication with the first valve and the second valve, the controller configured to coordinate movement of the working piston and the intensification piston between a rest state, an advancing state, an advanced state, an intensified state, and a retracted state, wherein the advanced state is configured to engage the end portion with a workpiece, and the intensified state is configured to perform an operation on the workpiece with the end portion.
